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Experiment No 11

Aim: Write queries to sort and aggregate the data in a table using HiveQL

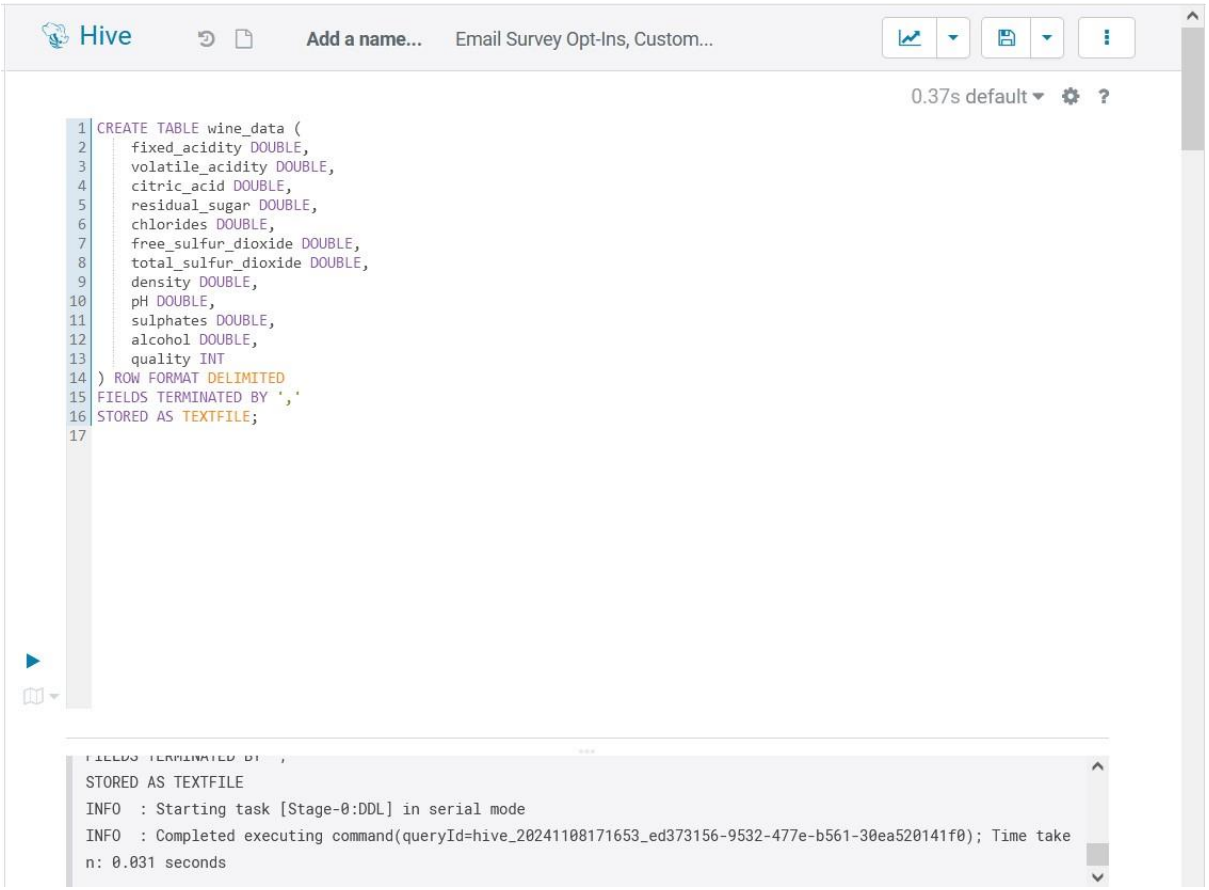
Objective:

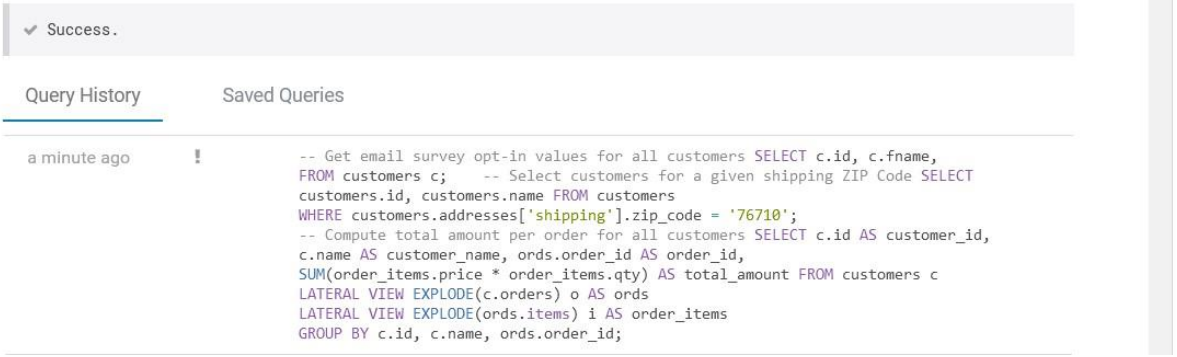
The objective of this experiment is to demonstrate how to write queries to sort and aggregate data stored in a Hive table using HiveQL.

Tools and Technologies:

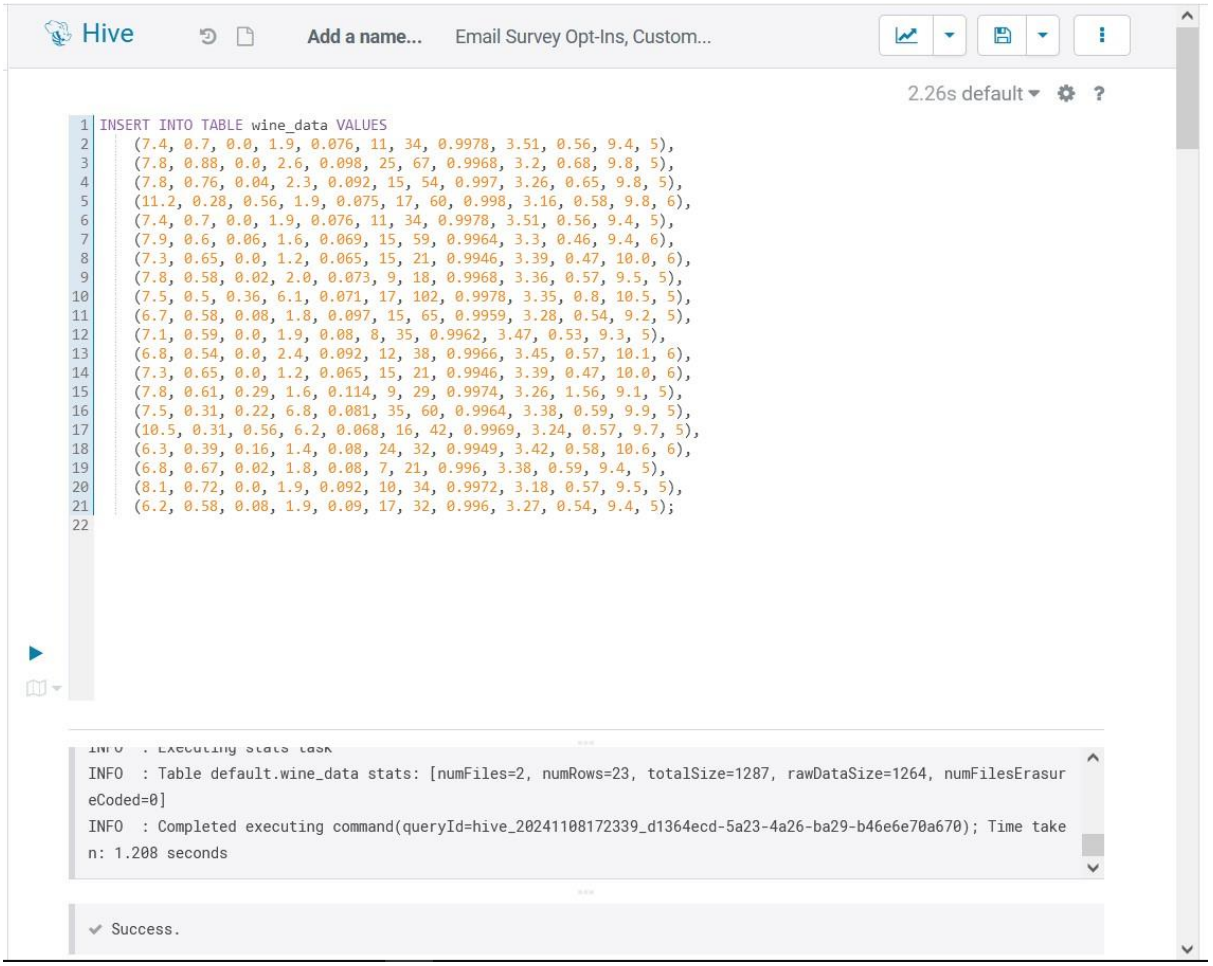
- Hive: A data warehouse infrastructure built on top of Hadoop for providing data summarization, query, and analysis.
- HiveQL: Hive Query Language, which is similar to SQL and used for querying and managing data in Hive.

1.Creating the Table in Hive

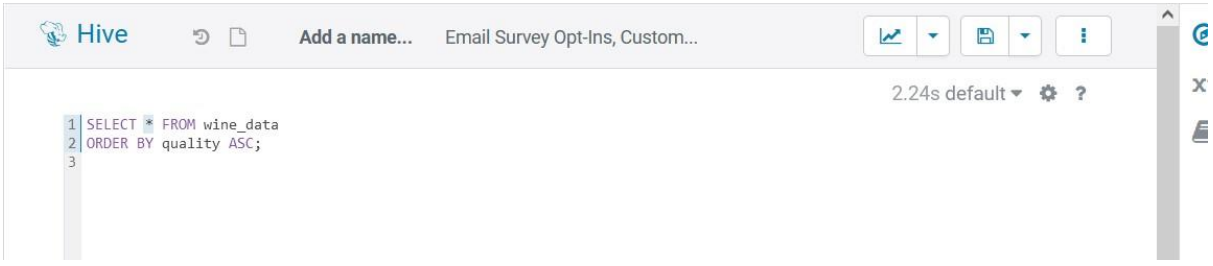


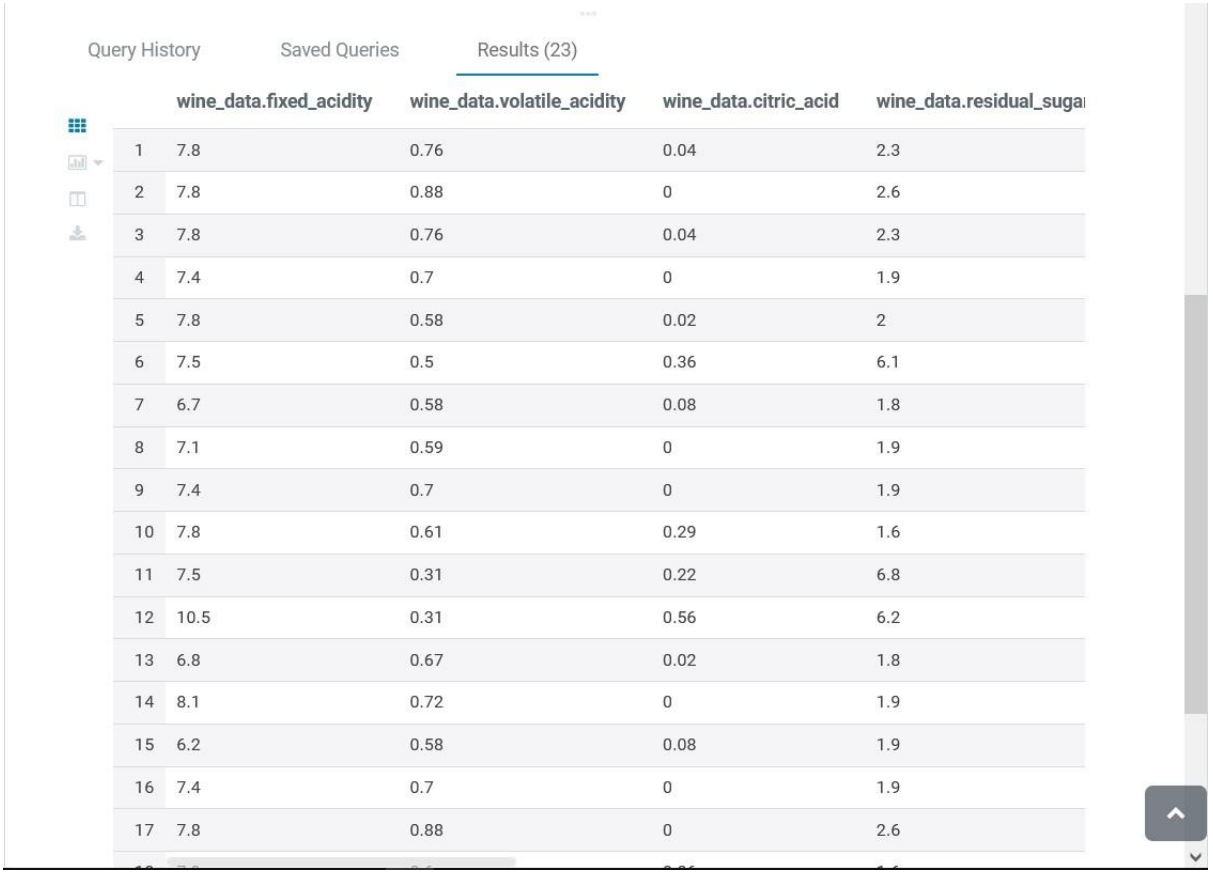


2. Inserting data into Hive



3. Sorting Data in the Hive Table

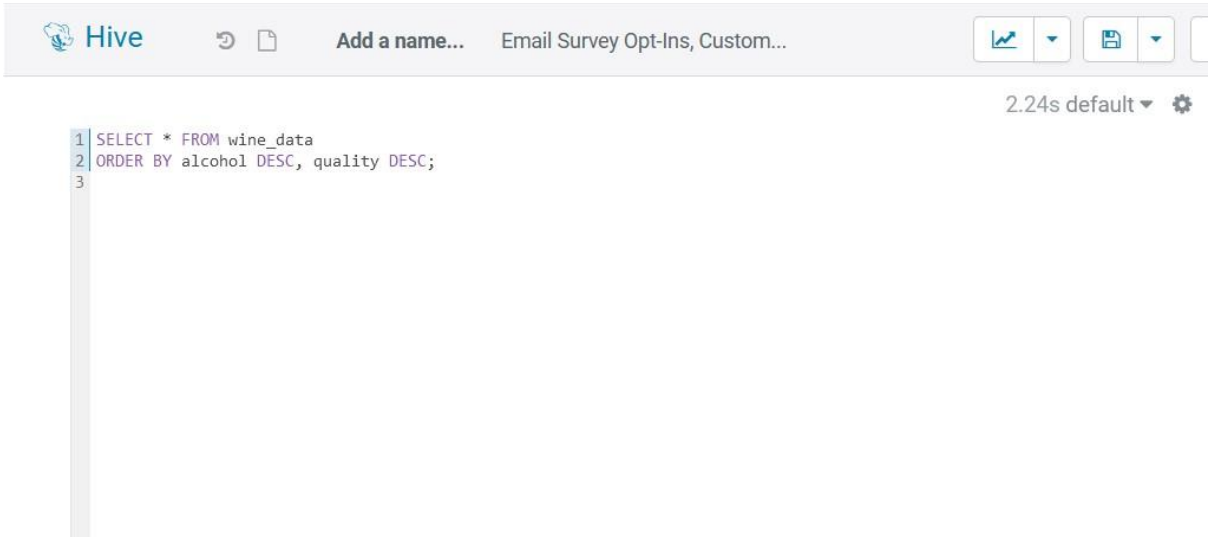




Query History | Saved Queries | Results (23)

	wine_data.fixed_acidity	wine_data.volatile_acidity	wine_data.citric_acid	wine_data.residual_sugar
1	7.8	0.76	0.04	2.3
2	7.8	0.88	0	2.6
3	7.8	0.76	0.04	2.3
4	7.4	0.7	0	1.9
5	7.8	0.58	0.02	2
6	7.5	0.5	0.36	6.1
7	6.7	0.58	0.08	1.8
8	7.1	0.59	0	1.9
9	7.4	0.7	0	1.9
10	7.8	0.61	0.29	1.6
11	7.5	0.31	0.22	6.8
12	10.5	0.31	0.56	6.2
13	6.8	0.67	0.02	1.8
14	8.1	0.72	0	1.9
15	6.2	0.58	0.08	1.9
16	7.4	0.7	0	1.9
17	7.8	0.88	0	2.6

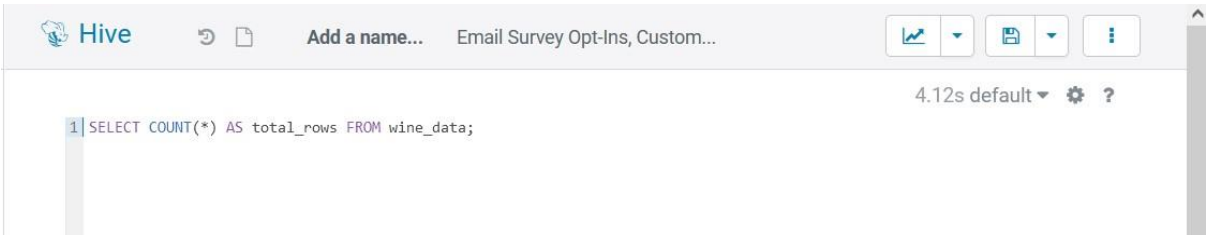
Sort in Descending Order by Multiple Columns (`alcohol` and `quality`):



Query History | Saved Queries | Results (23)

	wine_data.fixed_acidity	wine_data.volatile_acidity	wine_data.citric_acid	wine_data.residual_sugar
1	6.3	0.39	0.16	1.4
2	7.5	0.5	0.36	6.1
3	6.8	0.54	0	2.4
4	7.3	0.65	0	1.2
5	7.3	0.65	0	1.2
6	7.5	0.31	0.22	6.8
7	11.2	0.28	0.56	1.9
8	7.8	0.76	0.04	2.3
9	7.8	0.88	0	2.6
10	7.8	0.76	0.04	2.3
11	7.8	0.88	0	2.6
12	10.5	0.31	0.56	6.2
13	7.8	0.58	0.02	2
14	8.1	0.72	0	1.9
15	7.9	0.6	0.06	1.6
16	7.4	0.7	0	1.9
17	6.8	0.67	0.02	1.8

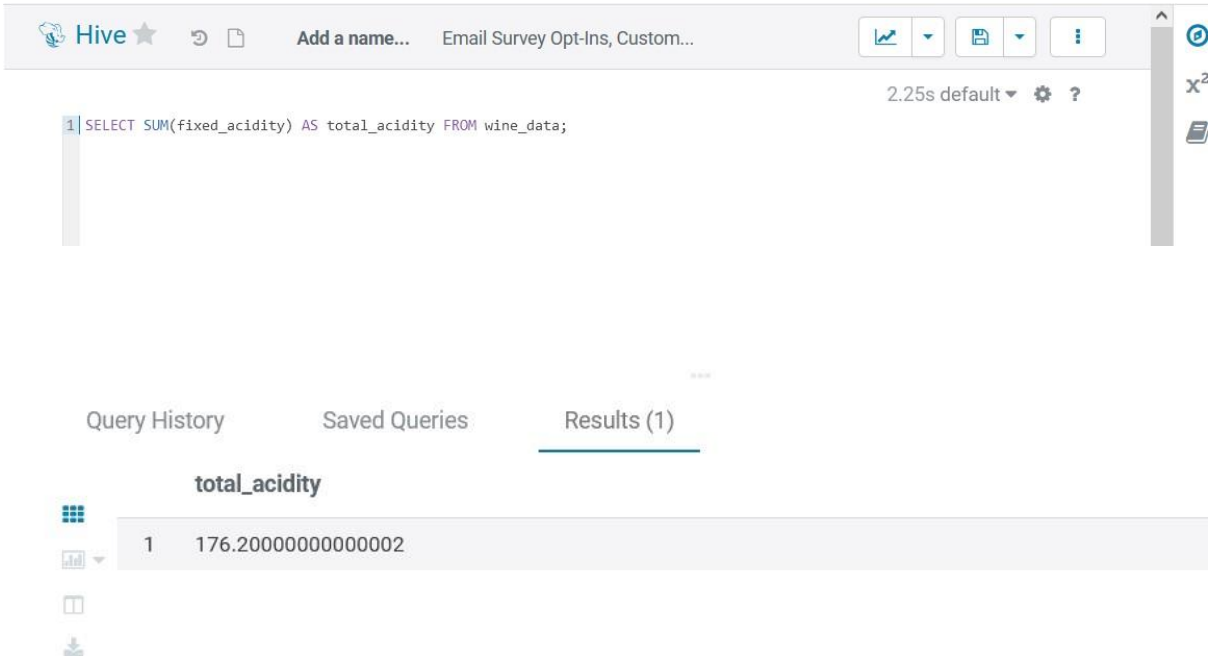
4. Aggregating Data in the Hive Table



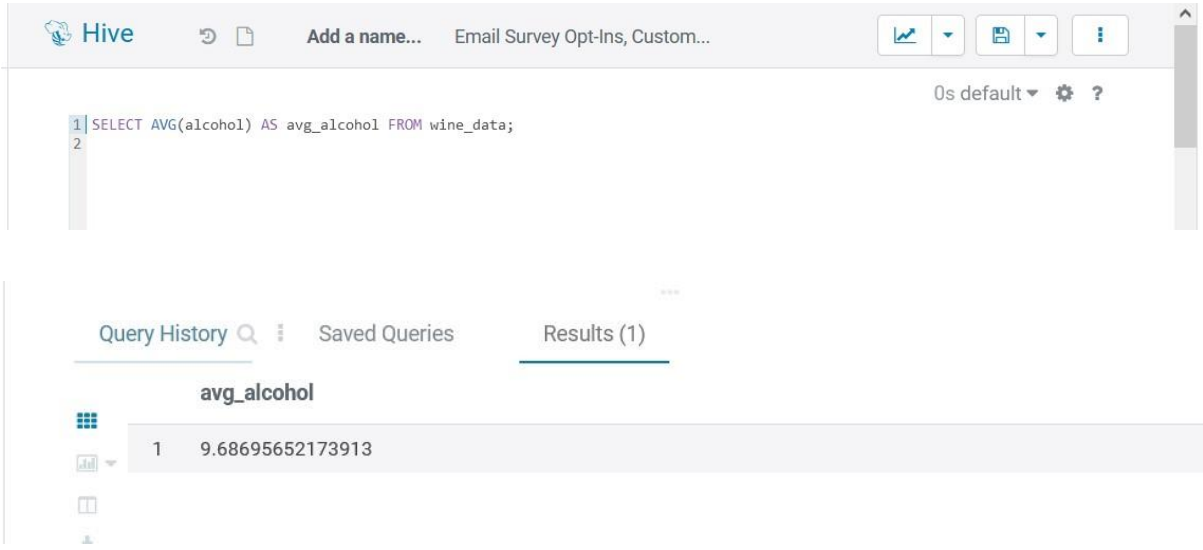
Query History | Saved Queries | Results (1)

	total_rows
1	23

Calculate the Sum of a Column (`fixed_acidity`):

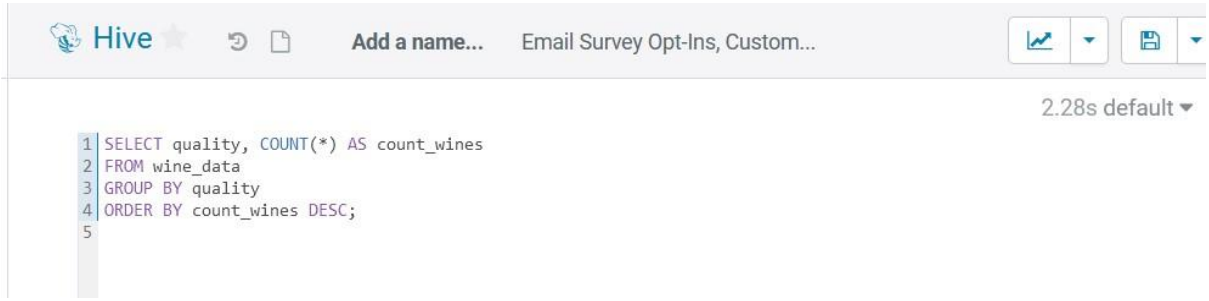


Find the Average of a Column (`alcohol`):

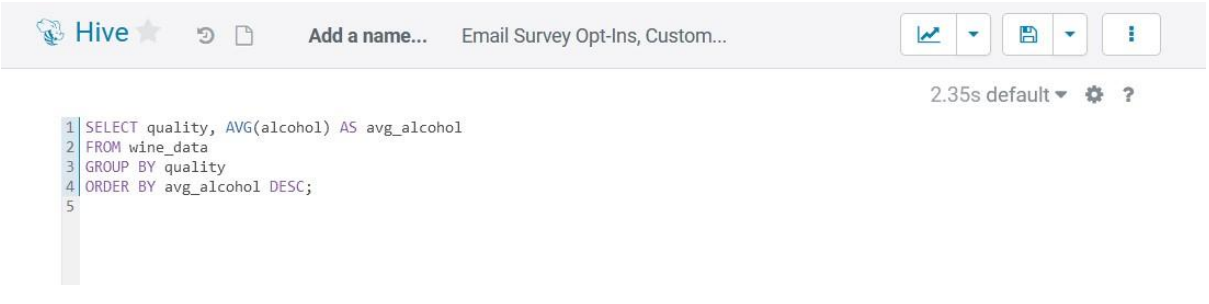


5. Combining Sorting and Aggregation

Count Records Grouped by `quality` and Sort by Count in Descending Order:



Calculate Average `alcohol` Content per `quality` and Sort by Average in Descending Order:



POSTLAB:

1.Explain the difference between Managed and External tables in HiveQL.

Ans:

Managed Table

- Hive manages the table data and metadata.
- Data is stored in a default location within Hive's warehouse directory (e.g., `/user/hive/warehouse`).
- When you drop a managed table, both data and metadata are deleted.
- Suitable for temporary or intermediate data where Hive should manage data lifecycle.

External Table

- Hive manages only the metadata, not the data itself.
- Data can be stored in any specified location, like HDFS or external storage.
- When you drop an external table, only the metadata is deleted; the data remains intact.
- Ideal for shared or persistent data that might be accessed by other applications outside Hive.

2. How do you optimize HiveQL queries?

Ans:

Partitioning:

- Split large tables into partitions based on columns (e.g., date) to reduce the amount of data scanned.

Bucketing:

- Group data within partitions into buckets, which improves performance for joins and sampling.

Use Appropriate File Formats:

- Use columnar formats like ORC or Parquet for better compression and faster reads.

Compression:

- Enable compression for intermediate data to reduce data transfer time.

Avoid `SELECT *`:

- Select only necessary columns to minimize data read and processed.

Use Vectorized Query Execution:

- Enable vectorization for certain operators to process batches of rows, which speeds up operations.

Optimize Joins:

- Use map-side joins for small tables and sort-merge joins for large tables.

Increase Parallelism:

- Adjust configuration to increase the number of mappers and reducers for better parallel processing.